CS4423 - Networks

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2. Centrality Measures

Lecture 8: Degree and Eigenvector Centrality

What is it that makes a node in a network important?

Key actors in a social network can be identified through **centrality measures**. The question of what it means to be central has a number of different answers. Accordingly, in the context of social network analysis, a variety of different centrality measures have been developed.

Here we introduce, in addition to the **degree centrality** that we have already seen, three more further measures:

- **eigenvector centrality**, defined in terms of properties of the network's **adjacency matrix**,
- **closeness centrality**, defined in terms of a nodes **distance** to other nodes on the network,
- **betweenness centrality**, defined in terms of **shortest paths**.

Start by importing the necessary `python` libraries into this jupyter notebook. (Actually, `networkx` interacts with a number of useful `python` libraries, some of which are loaded automatically, while others have to be `import` ed explicitly, depending on the circumstances. In the following, we will make explicit use of `Pandas` and `Numpy` .)

In [1]:
```
import networkx as nx
import numpy as np
import pandas as pd
opts = { "with_labels": True, "node_color": 'y'}
```

Degree Centrality

The **degree** of a node is its number of neighbors in the graph. This number can serve as a simple measure of centrality.

Consider the following network of **florentine families**, linked by **marital ties**.

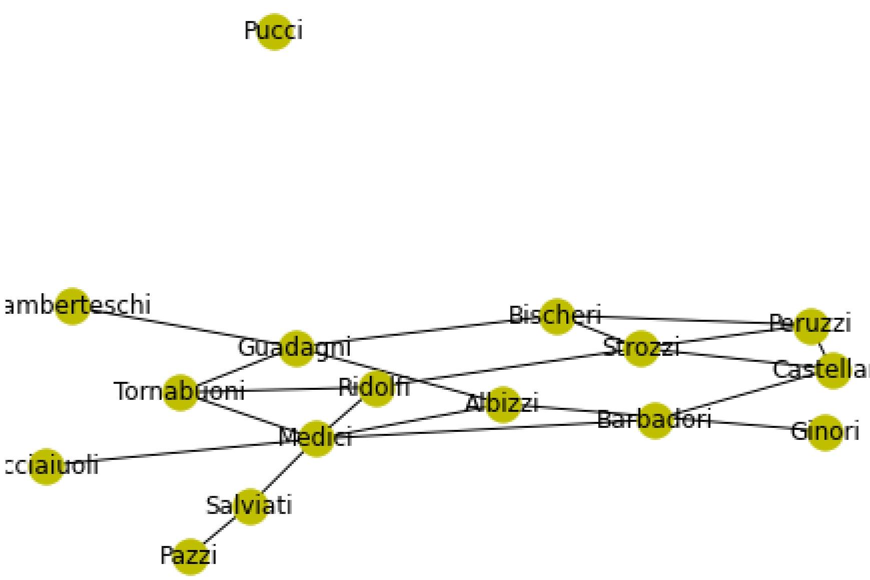
In [2]:
```
G = nx.florentine_families_graph()
list(G.nodes())
```

Out[2]:
```
['Acciaiuoli',
 'Medici',
 'Castellani',
 'Peruzzi',
 'Strozzi',
 'Barbadori',
 'Ridolfi',
 'Tornabuoni',
 'Albizzi',
 'Salviati',
 'Pazzi',
 'Bischeri',
 'Guadagni',
 'Ginori',
 'Lamberteschi']
```

Unfortunately, this version of the graph misses the isolated node `'Pucci'` of the original graph. Let's just add it and draw the resulting graph.

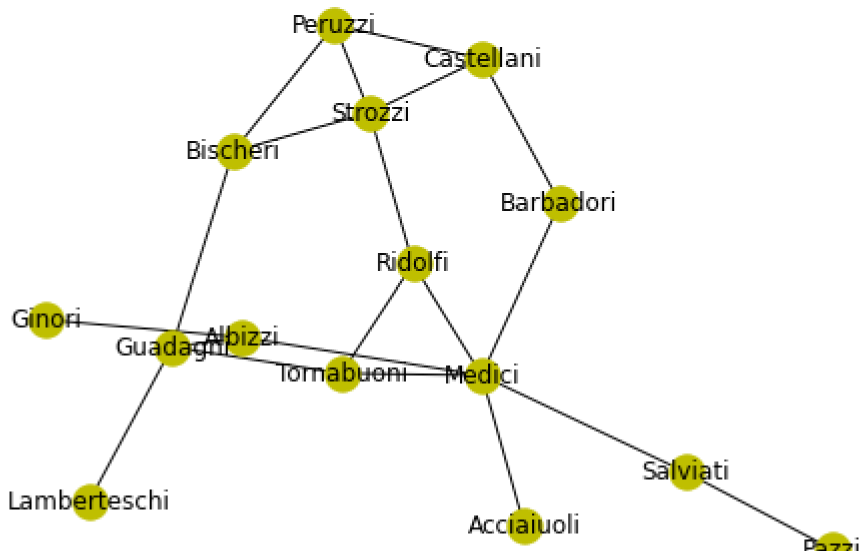
In [3]:
```
cc = list(G.nodes())
G.add_node('Pucci')
```

In [4]:
```
nx.draw(G, **opts)
```



The large connected component can be constructed and drawn as a subgraph. Let's call it `GG`.

In [5]:
```
GG = G.subgraph(cc)
nx.draw(GG, **opts)
```



Known indicators of the importance of these families are their **wealth**, and the number of seats on the city council (**priorates**). These measures can be compared with the node degree in the graph `G` .

In [6]:
```
wealth = {
  'Acciaiuoli': 10, 'Albizzi': 36, 'Barbadori': 55, 'Bischeri': 44,
  'Castellani': 20, 'Ginori': 32, 'Guadagni': 8, 'Lamberteschi': 42,
  'Medici': 103, 'Pazzi': 48, 'Peruzzi': 49, 'Pucci': 3,
  'Ridolfi': 27, 'Salviati': 10, 'Strozzi': 146, 'Tornabuoni': 48,
}

priorates = {
  'Acciaiuoli': 53, 'Albizzi': 65, 'Barbadori': 'n/a', 'Bischeri': 12,
  'Castellani': 22, 'Ginori': 'n/a', 'Guadagni': 21, 'Lamberteschi': 0,
  'Medici': 53, 'Pazzi': 'n/a', 'Peruzzi': 42, 'Pucci': 0,
  'Ridolfi': 38, 'Salviati': 35, 'Strozzi': 74, 'Tornabuoni': 'n/a',
}
```

In [7]:
```
nx.set_node_attributes(G, wealth, 'wealth')
nx.set_node_attributes(G, priorates, 'priorates')
nx.set_node_attributes(G, dict(G.degree()), 'degree')
```

In [8]:
```
print(dict(G.degree()))
```

```
{'Acciaiuoli': 1, 'Medici': 6, 'Castellani': 3, 'Peruzzi': 3, 'Strozz
i': 4, 'Barbadori': 2, 'Ridolfi': 3, 'Tornabuoni': 3, 'Albizzi': 3, 'S
alviati': 2, 'Pazzi': 1, 'Bischeri': 3, 'Guadagni': 4, 'Ginori': 1, 'L
amberteschi': 1, 'Pucci': 0}
```

In [9]:
```
G.nodes['Pazzi']
```

Out[9]:
```
{'wealth': 48, 'priorates': 'n/a', 'degree': 1}
```

In [10]:
```
dict(G.nodes(data=True))
```

Out[10]:
```
{'Acciaiuoli': {'wealth': 10, 'priorates': 53, 'degree': 1},
 'Medici': {'wealth': 103, 'priorates': 53, 'degree': 6},
 'Castellani': {'wealth': 20, 'priorates': 22, 'degree': 3},
 'Peruzzi': {'wealth': 49, 'priorates': 42, 'degree': 3},
 'Strozzi': {'wealth': 146, 'priorates': 74, 'degree': 4},
 'Barbadori': {'wealth': 55, 'priorates': 'n/a', 'degree': 2},
 'Ridolfi': {'wealth': 27, 'priorates': 38, 'degree': 3},
 'Tornabuoni': {'wealth': 48, 'priorates': 'n/a', 'degree': 3},
 'Albizzi': {'wealth': 36, 'priorates': 65, 'degree': 3},
 'Salviati': {'wealth': 10, 'priorates': 35, 'degree': 2},
 'Pazzi': {'wealth': 48, 'priorates': 'n/a', 'degree': 1},
 'Bischeri': {'wealth': 44, 'priorates': 12, 'degree': 3},
 'Guadagni': {'wealth': 8, 'priorates': 21, 'degree': 4},
 'Ginori': {'wealth': 32, 'priorates': 'n/a', 'degree': 1},
 'Lamberteschi': {'wealth': 42, 'priorates': 0, 'degree': 1},
 'Pucci': {'wealth': 3, 'priorates': 0, 'degree': 0}}
```

In [11]:
```
pd.DataFrame.from_dict(
    dict(G.nodes(data=True)),
    orient='index'
).sort_values('degree', ascending=False)
```

Out[11]:

	wealth	priorates	degree
Medici	103	53	6
Strozzi	146	74	4
Guadagni	8	21	4
Castellani	20	22	3
Peruzzi	49	42	3
Ridolfi	27	38	3
Tornabuoni	48	n/a	3
Albizzi	36	65	3
Bischeri	44	12	3
Barbadori	55	n/a	2
Salviati	10	35	2
Acciaiuoli	10	53	1
Pazzi	48	n/a	1
Ginori	32	n/a	1
Lamberteschi	42	0	1
Pucci	3	0	0

Definition (Degree Centrality). In a (simple) graph $G = (X, E)$, with $X = \{0, \dots, n-1\}$ and adjacency matrix $A = (a_{ij})$, the **degree centrality** c_i^D of node $i \in X$ is defined as

$$c_i^D = k_i = \sum_j a_{ij},$$

where k_i is the degree of node i.

The **normalized degree centrality** C_i^D of node $i \in X$ is defined as

$$C_i^D = \frac{k_i}{n-1} = \frac{c_i^D}{n-1},$$

the degree centrality of node i divided by its potential number of neighbors in the graph.

In a directed graph one distinguishes between the **in-degree** and the **out-degree** of a node and defines the **in-degree centrality** and the **out-degree centrality** accordingly.

In [12]:
```
nx.set_node_attributes(G, nx.degree_centrality(G), '$C_i^D$')
```

In [13]:
```
pd.DataFrame.from_dict(
    dict(G.nodes(data=True)),
    orient='index'
).sort_values('degree', ascending=False)
```

Out[13]:

	wealth	priorates	degree	C_i^D
Medici	103	53	6	0.400000
Strozzi	146	74	4	0.266667
Guadagni	8	21	4	0.266667
Castellani	20	22	3	0.200000
Peruzzi	49	42	3	0.200000
Ridolfi	27	38	3	0.200000
Tornabuoni	48	n/a	3	0.200000
Albizzi	36	65	3	0.200000
Bischeri	44	12	3	0.200000
Barbadori	55	n/a	2	0.133333
Salviati	10	35	2	0.133333
Acciaiuoli	10	53	1	0.066667
Pazzi	48	n/a	1	0.066667
Ginori	32	n/a	1	0.066667
Lamberteschi	42	0	1	0.066667
Pucci	3	0	0	0.000000

Eigenvectors and Centrality

Recall from **Linear Algebra** that a (n-dimensional) vector v is called an **eigenvector** of a square ($n \times n$)-matrix A, if

$$Av = \lambda v$$

for some scalar (number) λ (which is then called an **eigenvalue** of the matrix A)

The basic idea of eigenvector centrality is that **a node's ranking in a network should relate to the rankings of the nodes it is connected to**. More specifically, up to some scalar λ, the centrality c_i^E of node i should be equal to the sum if the centralities c_j^E of its neighbor nodes j. In terms of the adjacency matrix $A = (a_{ij})$, this relationship is expressed as

$$\lambda c_i^E = \sum_j a_{ij} c_j^E,$$

which in turn, in matrix language is

$$\lambda c^E = Ac_E,$$

for the vector $c^E = (c_i^E)$, which then is an eigenvector of A.

How to find c^E? Or λ? Linear Algebra:

1. Find the *characteristic polynomial* $p_A(x)$ of A (as *determinant* of the matrix $xI - A$, where I is the $n \times n$-identity matrix);
2. Find the *roots* λ of $p_A(x)$ (i.e. scalars λ such that $p_A(\lambda) = 0$;
3. Find a *nontrivial solution* of the linear system $(\lambda I - A)c = 0$ (where 0 stands for an all-0 column vector, and $c = (c_1, \ldots, c_n)$ is a column of *unknowns*).

In [14]:
```
A = nx.adjacency_matrix(G).toarray()
A
```

Out[14]:
```
array([[0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [1, 0, 0, 0, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0],
       [0, 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0],
       [0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0],
       [0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0],
       [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0],
       [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 1, 0, 0, 1, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0],
       [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

In `numpy`, the function `poly` finds the characteristic polynomial of a matrix, for example of the 2×2 matrix

$$B = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$$

In [15]:
```
B = np.array([[2,2],[3,1]])
poly = np.poly(B)
```

In [16]:
```
print(poly)
```

```
[ 1. -3. -4.]
```

Thus $p_B(x) = x^2 - 3x - 4$.

The eigenvalues and eigenvectors of B are found by `np.linalg.eig`:

In [17]:
```
l, v = np.linalg.eig(B)
vv = v.transpose()
```

2 Eigenvalues:

In [18]:
```
print(l);
```

```
[ 4. -1.]
```

and 2 eigenvectors (the rows of this matrix):

In [19]:
```
print (vv);
```

```
[[ 0.70710678  0.70710678]
 [-0.5547002   0.83205029]]
```

Check: does $B \cdot v_0 = l_0 \, v_0$?

In [20]:
```
print(l[0]*vv[0])
```

```
[2.82842712 2.82842712]
```

In [21]:
```
print(np.matmul(B, vv[0]))
```

```
[2.82842712 2.82842712]
```

Now for the bigger matrix:

In [22]:
```
print(A)
```

```
[[0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0]
 [1 0 0 0 0 1 1 1 1 1 0 0 0 0 0 0]
 [0 0 0 1 1 1 0 0 0 0 0 0 0 0 0 0]
 [0 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0]
 [0 0 1 1 0 0 1 0 0 0 0 1 0 0 0 0]
 [0 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0]
 [0 1 0 0 1 0 0 1 0 0 0 0 0 0 0 0]
 [0 1 0 0 0 0 1 0 0 0 0 0 1 0 0 0]
 [0 1 0 0 0 0 0 0 0 0 0 0 1 1 0 0]
 [0 1 0 0 0 0 0 0 0 0 1 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0 1 0 0 0 0 0 0]
 [0 0 0 1 1 0 0 0 0 0 0 0 1 0 0 0]
 [0 0 0 0 0 0 0 1 1 0 0 1 0 0 1 0]
 [0 0 0 0 0 0 0 0 1 0 0 0 0 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0]
 [0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0]]
```

Find its characteristic polynomial ...

In [23]:
```
np.poly(A)
```

Out[23]:
```
array([ 1.00000000e+00,  1.11022302e-15, -2.00000000e+01, -6.00000000e
+00,
        1.39000000e+02,  6.80000000e+01, -4.17000000e+02, -2.42000000e
+02,
        5.65000000e+02,  3.44000000e+02, -3.44000000e+02, -2.08000000e
+02,
        8.20000000e+01,  4.60000000e+01, -5.00000000e+00, -2.00000000e
+00,
        0.00000000e+00])
```

... and its eigenvalues ...???

Numerical Linerar Algebra: forget algebraic precision, use the **Power method**:

1. start with $u = (1, 1, \dots, 1)$, say;
2. keep replacing $u \leftarrow Au$ until $u/\|u\|$ becomes stable ...

If A has a *dominant* eigenvalue λ_0 then u will *converge* to an eigenvector for the eigenvalue λ_0.

Restore the matrix A:

In [24]:
```
A = nx.adjacency_matrix(G)
```

Provide an all-1 vector u and normalize it:

In [25]:
```
u = [1 for x in A]
print(u)
print(u/np.linalg.norm(u))
```

```
[1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1]
[0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25
 0.25 0.25]
```

Multiply by A:

In [26]:
```
v = A * u
print(v)
print(v/np.linalg.norm(v))
```

```
[1 6 3 3 4 2 3 3 3 2 1 3 4 1 1 0]
[0.08638684 0.51832106 0.25916053 0.25916053 0.34554737 0.17277369
 0.25916053 0.25916053 0.25916053 0.17277369 0.08638684 0.25916053
 0.34554737 0.08638684 0.08638684 0.        ]
```

Repeat this a couple of times.

In [27]:
```
for i in range(40):
    u = A * u
    u = u/np.linalg.norm(u)
```

Now, u should have stabilized and a further multiplication by A should be the same as multiplying u with the eigenvalue l: $Au = lu$.

In [28]:
```
print("u = ", u)
v = A * u
v = v/np.linalg.norm(v)
print("v = ", v)
```

```
u =  [0.13217021 0.43026554 0.25902292 0.27572882 0.355973   0.2117222
6
 0.34156271 0.32586538 0.24398716 0.14593566 0.04480665 0.282814
 0.2890864  0.07491127 0.08880275 0.        ]
v =  [0.13214123 0.43034359 0.25902863 0.27573136 0.35598634 0.2116911
9
 0.34154438 0.32582335 0.24393056 0.14590205 0.04481911 0.28278835
 0.28913983 0.07493225 0.08878292 0.        ]
```

The stability of u can be tested by computing the norm of the difference of u and v.

In [29]:
```
v = v/np.linalg.norm(v)
print("||v - u|| = ", np.linalg.norm(v - u))
```

```
||v - u|| =  0.0001380299403227346
```

So l can be found as the quotient of two corresponding vector entries

In [30]:
```
v = A * u
l = v[2]/u[2]
print("l = ", l)
```

```
l =  3.2561754813946275
```

The function `np.linalg.eig` does all of this and more. It finds **all** eigenvalues of the adjaceny matrix A, and a list `w` of eigenvectors.

The first eigenvalue in the list `l` below should be the one that the power method discovered.

The entries in the corresponding eigenvector are then the eigenvector centralities of the nodes in the graph.

In [31]:
```
l, w = np.linalg.eig(A.toarray())
print(l)
print(w[:,0])
```

```
[ 3.25610375  2.42381396 -2.69583872  1.70899078 -2.06786891 -1.870722
41
  1.05403772  0.93839893  0.60199089  0.25781512 -0.20243481 -1.193293
97
 -0.57626063 -0.76538496 -0.86934672  0.        ]
[0.13215429 0.43030809 0.25902617 0.27573037 0.35598045 0.21170525
 0.34155264 0.3258423  0.24395611 0.1459172  0.04481344 0.28280009
 0.2891156  0.07492271 0.08879189 0.        ]
```

The `networkx` package computes them as follows:

In [32]:
```
eigen_cen = nx.eigenvector_centrality(G)
eigen_cen
```

Out[32]:
```
{'Acciaiuoli': 0.13215731952853418,
 'Medici': 0.4303154258349922,
 'Castellani': 0.2590200378423514,
 'Peruzzi': 0.2757224374104833,
 'Strozzi': 0.355973032646045,
 'Barbadori': 0.21170574706479847,
 'Ridolfi': 0.34155442590743645,
 'Tornabuoni': 0.32584670416957395,
 'Albizzi': 0.24396052967544765,
 'Salviati': 0.1459208416417183,
 'Pazzi': 0.04481493970386308,
 'Bischeri': 0.2827943958713356,
 'Guadagni': 0.2891171573226501,
 'Ginori': 0.0749245316027793,
 'Lamberteschi': 0.08879253113499551,
 'Pucci': 1.5210930780184965e-24}
```

The theoretical foundation for this approach is provided by the following Linear Algebra [theorem (https://en.wikipedia.org/wiki/Perron%E2%80%93Frobenius_theorem)](https://en.wikipedia.org/wiki/Perron%E2%80%93Frobenius_theorem) from 1907/1912.

> **Theorem.** (Perron-Frobenius for irreducible matrices.) Suppose that A is a square, nonnegative, irreducible matrix. Then:
>
> - A has a real eigenvalue $\lambda > 0$ with $\lambda \geq |\lambda'|$ for all eigenvalues λ' of A;
> - λ is a simple root of the characteristic polynomial of A;
> - there is a λ-eigenvector v with $v > 0$.

Here, a matrix A is called **reducible** if, for some simultaneous permutation of its rows and columns, it has the block form

$$A = \begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}.$$

And A is **irreducible** if it is not reducible.

The incidence matrix of a simple graph G is irreducible if and only if G is connected.

Example. Recall that
$$\begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \end{pmatrix} = 4 \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$
making the vector $\binom{1}{1}$ an **eigenvector** for the **eigenvalue** $\lambda = 4$ of the matrix A.

In this example

- all entries a_{ij} of the matrix $A = (a_{ij})$ are positive;
- the eigenvalue 4 is strictly larger than the magnitude $|\lambda'|$ of all the other (complex or real) eigenvalues of A (here, $\lambda' = -1$);
- and the eigenvalue $\lambda = 4$ has an eigenvector with all its entries positive.

The Perron-Frobenius Theorem states that the above observations are **no coincidence**.

Definition (Eigenvector centrality). In a simple, connected graph G, the **eigenvector centrality** c_i^E of node i is defined as
$$c_i^E = u_i,$$
where $u = (u_1, \dots, u_n)$ is the (unique) normalized eigenvector of the adjacency matrix A of G with eigenvalue λ, and where $\lambda > |\lambda'|$ for all eigenvalues λ' of A.

The **normalised eigenvector centrality** of node i is defined as
$$C_i^E = \frac{c_i^E}{C^E},$$
where $C^E = \sum_j c_j^E$.

Let's attach the eigenvector centralities as node attributes and display the resulting table.

In [33]:
```
eigen_cen = nx.eigenvector_centrality(G)
nx.set_node_attributes(G, eigen_cen, '$c_i^E$')
```

In [34]:
```
pd.DataFrame.from_dict(
    dict(G.nodes(data=True)),
    orient='index'
).sort_values('degree', ascending=False)
```

Out[34]:

	wealth	priorates	degree	C_i^D	c_i^E
Medici	103	53	6	0.400000	4.303154e-01
Strozzi	146	74	4	0.266667	3.559730e-01
Guadagni	8	21	4	0.266667	2.891172e-01
Castellani	20	22	3	0.200000	2.590200e-01
Peruzzi	49	42	3	0.200000	2.757224e-01
Ridolfi	27	38	3	0.200000	3.415544e-01
Tornabuoni	48	n/a	3	0.200000	3.258467e-01
Albizzi	36	65	3	0.200000	2.439605e-01
Bischeri	44	12	3	0.200000	2.827944e-01
Barbadori	55	n/a	2	0.133333	2.117057e-01
Salviati	10	35	2	0.133333	1.459208e-01
Acciaiuoli	10	53	1	0.066667	1.321573e-01
Pazzi	48	n/a	1	0.066667	4.481494e-02
Ginori	32	n/a	1	0.066667	7.492453e-02
Lamberteschi	42	0	1	0.066667	8.879253e-02
Pucci	3	0	0	0.000000	1.521093e-24

Save the graph and its attributes for future use.

In [35]:
```
nx.write_yaml(G, "data/florentine.yml")
```

Code Corner

`numpy`

- `array` : [doc] (https://numpy.org/doc/stable/reference/routines.array-creation.html) `numpy` array creation
- `poly` : [doc] (https://numpy.org/doc/stable/reference/generated/numpy.poly.html)
- `matmul` : [doc] (https://numpy.org/doc/stable/reference/generated/numpy.matmul.html)
- `linalg.eig` : [doc] (https://numpy.org/doc/stable/reference/generated/numpy.linalg.eig.html)

- `linalg.norm` : [doc] (https://numpy.org/doc/stable/reference/generated/numpy.linalg.norm.html)

`pandas`

- `DataFrame.from_dict` : [doc] (https://pandas.pydata.org/pandas-docs/stable/reference/api/pandas.DataFrame.from_dict.html)

`networkx`

- `florentine_families_graph` : [doc] (https://networkx.github.io/documentation/stable/reference/generators.html#module-networkx.generators.social) the marital links between powerful families in 15th century Florence (excluding the isolated Pucci family).
- `set_node_attributes` : [doc] (https://networkx.github.io/documentation/stable/reference/generated/networkx.classes.functi set a named attribute for several/all nodes in a graph.
- `degree_centrality` : [doc] (https://networkx.github.io/documentation/stable/reference/algorithms/generated/networkx.alg
- `eigenvector_centrality` : [doc] (https://networkx.github.io/documentation/stable/reference/algorithms/generated/networkx.alg
- `write_yaml` : [doc] (https://networkx.github.io/documentation/stable/reference/readwrite/yaml.html) store a graph on file in YAML format.

Exercises

1. Apply the power method to the 2×2 matrix
$$B = \begin{pmatrix} 2 & 2 \\ 3 & 1 \end{pmatrix}$$
from above. Which eigenvalue and which eigenvector can be found in this way?
2. Compute the characteristic polynomial and the eigenvalues of the matrix
$$M = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

3. Determine the degree centrality and the eigenvector centrality of the nodes in some random trees. What do you observe?
4. Compute the eigenvector and degree centralities of the nodes of the Petersen graph. What do you observe?

In [ ]:

```

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